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Last login: Wed Aug 12 14:01:10 on tty807
amrutha@amruthas-MacBook-Air: ~ % ls -l /etc/passwd
-rw-r--r-- 1 root root 128 Aug 12 14:01 /etc/passwd
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
mongod 13725 amrutha 8u IPv4 6x21e7be7326d6161b 0t0 TCP localhost:27017 (LISTEN)
amrutha@amruthas-MacBook-Air: ~ % mongosh
Current Mongosh Log ID: 6a7c380c5aca84086fb2adac1
Connecting to: mongod://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.9.2
Using MongoDB: 8.3.7
Using Mongosh: 2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

The server generated these startup warnings when booting
2026-08-12T14:02:35.960+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-08-12T14:02:35.960+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip <address> to specify which IP addresses it should
serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning
2026-08-12T14:02:35.960+05:30: Soft rlimits for open file descriptors too low

test> use bookflow_db
switched to db bookflow_db
bookflow_db> db
bookflow_db> db.createCollection("book_metadata")
{ ok: 1 }
bookflow_db> show collections
book_metadata
bookflow_db> db.book_metadata.countDocuments()
3
bookflow_db> db.book_metadata.find()
[
  {
    _id: 0, title: 1, isbn: 1
  },
  {
    title: 'Python Programming',
    title: 'Database Systems',
    title: 'Machine Learning'
  }
]
bookflow_db> db.book_metadata.insertOne({
  title: 'Digital Electronics',
  isbn: 'ISBN000',
  published_year: 2019,
  format: 'Digital PDF',
  technical_specs: {
    pages: 380,
    language: 'English',
    edition: '4th'
  },
  reviews: [
    {
      member_id: 107,
      rating: 4,
      comment: 'Good introduction to electronics'
    },
    {
      member_id: 108,
      rating: 3,
      comment: 'Some topics are difficult'
    }
  ]
})

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    }
  }
  acknowledged: true,
  insertedId: ObjectId('6a7c380c5aca84086fb2adac2')
}
bookflow_db> db.book_metadata.countDocuments()
4
bookflow_db> db.book_metadata.find().pretty()
{
  _id: ObjectId('6a705f5589ee399343808289'),
  book_id: 1,
  title: 'Python Programming',
  member_id: 101,
  published_year: 2022,
  format: 'Digital',
  technical_specs: { language: 'English', pages: 450 },
  reviews: [
    { reviewer: 'Amrutha', rating: 5, comment: 'Excellent book' },
    { reviewer: 'Rahul', rating: 4, comment: 'Very useful' }
  ]
},
{
  _id: ObjectId('6a705f5589ee39934380828a'),
  book_id: 2,
  title: 'Database Systems',
  member_id: 102,
  published_year: 2021,
  format: 'Printed',
  technical_specs: { language: 'English', pages: 380 },
  reviews: [
    { reviewer: 'Rahul', rating: 5, comment: 'Great explanations' },
    { reviewer: 'Priya', rating: 4, comment: 'Easy to understand' }
  ]
},
{
  _id: ObjectId('6a705f5589ee39934380828b'),
  book_id: 3,
  title: 'Machine Learning',
  member_id: 103,
  published_year: 2024,
  format: 'Digital',
  technical_specs: { language: 'English', pages: 520 },
  reviews: [
    { reviewer: 'Amrutha', rating: 5, comment: 'Excellent examples' },
    { reviewer: 'Priya', rating: 5, comment: 'Highly recommended' }
  ]
},
{
  _id: ObjectId('6a7c380c5aca84086fb2adac2'),
  title: 'Digital Electronics',
  isbn: 'ISBN000',
  published_year: 2019,
  format: 'Digital PDF',
  technical_specs: { pages: 380, language: 'English', edition: '4th' },
  reviews: [
    {
      member_id: 107,
      rating: 4,

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        comment: 'Good introduction to electronics'
    },
    { member_id: 108, rating: 3, comment: 'Some topics are difficult' }
  ]
}
bookflow_db> db.book_metadata.find(
  {'reviews.rating': {'$gt': 4 }}
)
{
  '_id': ObjectId('6a705f5589ee39934388289'),
  'book_id': 1,
  'title': 'Python Programming',
  'member_id': 101,
  'published_year': 2022,
  'format': 'Digital',
  'technical_specs': { language: 'English', pages: 450 },
  'reviews': [
    { reviewer: 'Amrutha', rating: 5, comment: 'Excellent book' },
    { reviewer: 'Rahul', rating: 4, comment: 'Very useful' }
  ]
},
{
  '_id': ObjectId('6a705f5589ee3993438828a'),
  'book_id': 2,
  'title': 'Database Systems',
  'member_id': 102,
  'published_year': 2021,
  'format': 'Printed',
  'technical_specs': { language: 'English', pages: 380 },
  'reviews': [
    { reviewer: 'Rahul', rating: 5, comment: 'Great explanations' },
    { reviewer: 'Priya', rating: 4, comment: 'Easy to understand' }
  ]
},
{
  '_id': ObjectId('6a705f5589ee3993438828b'),
  'book_id': 3,
  'title': 'Machine Learning',
  'member_id': 103,
  'published_year': 2024,
  'format': 'Digital',
  'technical_specs': { language: 'English', pages: 520 },
  'reviews': [
    { reviewer: 'Amrutha', rating: 5, comment: 'Excellent examples' },
    { reviewer: 'Priya', rating: 5, comment: 'Highly recommended' }
  ]
}
}
bookflow_db> db.book_metadata.find(
  {'format': {'$in': ['Digital PDF', 'Print'] }}
)
{
  '_id': ObjectId('6a7c38c5aca84806fb2adac2'),
  'title': 'Digital Electronics',
  'isbn': 'ISBN0804',
  'published_year': 2019,
  'format': 'Digital PDF',

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  technical_specs: { pages: 380, language: 'English', edition: '4th' },
  reviews: [
    {
      member_id: 107,
      rating: 4,
      comment: 'Good introduction to electronics'
    },
    { member_id: 108, rating: 3, comment: 'Some topics are difficult' }
  ]
}
}
bookflow_db> db.book_metadata.find(
  {'format': {'$regex': 'Digital'}, '$options': 'i' })
{
  '_id': ObjectId('6a705f5589ee39934388289'),
  'book_id': 1,
  'title': 'Python Programming',
  'member_id': 101,
  'published_year': 2022,
  'format': 'Digital',
  'technical_specs': { language: 'English', pages: 450 },
  'reviews': [
    { reviewer: 'Amrutha', rating: 5, comment: 'Excellent book' },
    { reviewer: 'Rahul', rating: 4, comment: 'Very useful' }
  ]
},
{
  '_id': ObjectId('6a705f5589ee3993438828b'),
  'book_id': 3,
  'title': 'Machine Learning',
  'member_id': 103,
  'published_year': 2024,
  'format': 'Digital',
  'technical_specs': { language: 'English', pages: 520 },
  'reviews': [
    { reviewer: 'Amrutha', rating: 5, comment: 'Excellent examples' },
    { reviewer: 'Priya', rating: 5, comment: 'Highly recommended' }
  ]
},
{
  '_id': ObjectId('6a7c38c5aca84806fb2adac2'),
  'title': 'Digital Electronics',
  'isbn': 'ISBN0804',
  'published_year': 2019,
  'format': 'Digital PDF',
  'technical_specs': { pages: 380, language: 'English', edition: '4th' },
  reviews: [
    {
      member_id: 107,
      rating: 4,
      comment: 'Good introduction to electronics'
    },
    { member_id: 108, rating: 3, comment: 'Some topics are difficult' }
  ]
}
}
bookflow_db> db.book_metadata.aggregate([
  {

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    {
      $match: {
        published_year: { $gt: 2020 }
      }
    },
    {
      $sumwind: "$reviews"
    },
    {
      $group: {
        _id: "$title",
        average_rating: {
          $avg: "$reviews.rating"
        }
      }
    }
  })
}

{ _id: 'Database Systems', average_rating: 4.5 },
{ _id: 'Python Programming', average_rating: 4.5 },
{ _id: 'Machine Learning', average_rating: 5 }
}

bookflow_db> db.book_metadata.aggregate([
  {
    $match: {
      published_year: { $gt: 2020 }
    }
  },
  {
    $sumwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      average_rating: {
        $avg: "$reviews.rating"
      }
    }
  },
  {
    $sort: {
      average_rating: -1
    }
  }
])

{ _id: 'Machine Learning', average_rating: 5 },
{ _id: 'Python Programming', average_rating: 4.5 },
{ _id: 'Database Systems', average_rating: 4.5 }
}

bookflow_db> db.book_metadata.createIndex({
  "reviews.comment": "text"
})

reviews.comment_text
bookflow_db> db.book_metadata.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },

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    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  }
}

bookflow_db> db.book_metadata.find(
  {
    $text: {
      $search: "excellent"
    }
  }
)

[
  {
    _id: ObjectId('6a705f5589ee3993438828b'),
    book_id: 8,
    title: 'Machine Learning',
    member_id: 100,
    published_year: 2024,
    format: 'Digital',
    technical_specs: { language: 'English', pages: 520 },
    reviews: [
      { reviewer: 'Amrutha', rating: 5, comment: 'Excellent examples' },
      { reviewer: 'Priya', rating: 5, comment: 'Highly recommended' }
    ]
  },
  {
    _id: ObjectId('6a705f5589ee39934388289'),
    book_id: 9,
    title: 'Python Programming',
    member_id: 101,
    published_year: 2022,
    format: 'Digital',
    technical_specs: { language: 'English', pages: 450 },
    reviews: [
      { reviewer: 'Amrutha', rating: 5, comment: 'Excellent book' },
      { reviewer: 'Rahul', rating: 4, comment: 'Very useful' }
    ]
  }
]

bookflow_db> db.book_metadata.find(
  {
    $text: {
      $search: "excellent"
    }
  }
).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'excellent',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4f885230',
    planCacheShapeHash: '4f885230',

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planCacheKey: 'F34F3876',
optimizationTimeMillis: 0,
maxIndexedOrSolutionsReached: false,
maxIndexedAndSolutionsReached: false,
maxScansToSolveReached: false,
prunedSimilarIndexes: false,
winningPlan: {
  isCached: false,
  stage: 'TEXT_MATCH',
  nss: 'bookflow_db.book_metadata',
  indexPrefix: {},
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [ 'excel' ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    nss: 'bookflow_db.book_metadata',
    inputStage: {
      stage: 'IXSCAN',
      nss: 'bookflow_db.book_metadata',
      keyPattern: { _fts: 'text', _ftx: 1 },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnlower: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {}
    }
  },
  rejectedPlans: []
},
},
executionStats: {
  executionSuccess: true,
  nReturned: 2,
  executionTimeMillis: 1,
  totalKeysExamined: 2,
  totalDocsExamined: 2,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 2,
    executionTimeMillisEstimate: 0,
    works: 3,
    advanced: 2,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflow_db.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',

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    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [ 'excel' ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    },
    textIndexVersion: 3,
    docsRejected: 0,
    inputStage: {
      stage: 'FETCH',
      nReturned: 2,
      executionTimeMillisEstimate: 0,
      works: 3,
      advanced: 2,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'bookflow_db.book_metadata',
      docsExamined: 2,
      alreadyHasObj: 0,
      inputStage: {
        stage: 'IXSCAN',
        nReturned: 2,
        executionTimeMillisEstimate: 0,
        works: 3,
        advanced: 2,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        nss: 'bookflow_db.book_metadata',
        keyPattern: { _fts: 'text', _ftx: 1 },
        indexName: 'reviews.comment_text',
        isMultiKey: true,
        isUnlower: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'backward',
        indexBounds: {},
        keysExamined: 2,
        seeks: 1,
        dupesTested: 2,
        dupesDropped: 0,
        peakTrackedMemBytes: 18
      }
    },
    queryShapeHash: '5637AA384E2ADEAD97F86E65CDE99F8EE1FC651F238E25880AE18EA46711A',
    peakTrackedMemBytes: Long('18'),
    commands: {
      find: 'book_metadata',
      filter: { '$text': { '$search': 'excellent' } },
      '$db': 'bookflow_db'
    },
  },

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    },
    dupsDropped: 0,
    peakTrackedMemBytes: 18
  }
}
},
queryShapeHash: '5637A4A384E2ADEAD697F06E65CDEE9F8EE1FEC651F238E25B00AE18EA46711A',
peakTrackedMemBytes: Long('18'),
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'excellent' } },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'Amruthas-MacBook-Air.local',
  port: 27017,
  version: '8.3.7',
  gitVersion: '34eee04f34989abb7a3d91447976f033f4f74af2'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

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